

1.Employee Payroll System

```
import java.util.Scanner;

class Employee {

    int empID;
    String name;
    double basicSalary;
    double grossSalary;

    void inputDetail() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee ID: ");
        empID = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Name: ");
        name = sc.nextLine();

        System.out.print("Enter Basic Salary: ");
        basicSalary = sc.nextDouble();
    }

    void calculateSalary() {

        double HRA = basicSalary * 0.20;
        double DA = basicSalary * 0.10;

        grossSalary = basicSalary + HRA + DA;
    }

    void displayEmployee() {
        System.out.println("\nEmployee ID: " + empID);
        System.out.println("Name: " + name);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("Gross Salary: " + grossSalary);
    }
}

public class PayrollSystem {
    public static void main(String[] args) {

        Employee e1 = new Employee();
        Employee e2 = new Employee();

        e1.inputDetail();
        e1.calculateSalary();
        e1.displayEmployee();

        e2.inputDetail();
        e2.calculateSalary();
        e2.displayEmployee();
    }
}
```

2.Train Reservation System

```
class Train {

    int trainNumber;
    String source;
```

```

String destination;
int availableSeats;

void displayTrainInfo() {
    System.out.println("Train Number: " + trainNumber);
    System.out.println("Source: " + source);
    System.out.println("Destination: " + destination);
    System.out.println("Available Seats: " + availableSeats);
}

void bookTicket(int numSeats) {
    if (numSeats <= availableSeats) {
        availableSeats = availableSeats - numSeats;
        System.out.println("Ticket booked successfully");
    } else {
        System.out.println("Not enough seats available");
    }
}
}

public class TrainReservation {
    public static void main(String[] args) {

        Train t = new Train();

        t.trainNumber = 12345;
        t.source = "Nashik";
        t.destination = "Mumbai";
        t.availableSeats = 50;

        t.displayTrainInfo();

        t.bookTicket(5);

        t.displayTrainInfo();
    }
}

```

3. Library Book System

```

class Book {
    int bookID;
    String title;
    String author;
    boolean isAvailable = true;

    void issueBook() {
        if (isAvailable) {
            isAvailable = false;
            System.out.println("Book Issued");
        } else {
            System.out.println("Book Not Available");
        }
    }

    void returnBook() {
        isAvailable = true;
        System.out.println("Book Returned");
    }

    void displayBookInfo() {

```

```

        System.out.println(this.toString());
    }

    public String toString() {
        return "BookID: " + bookID +
            "\nTitle: " + title +
            "\nAuthor: " + author +
            "\nAvailable: " + isAvailable;
    }

    void searchBook(String keyword) {

        if (title.contains(keyword) || author.contains(keyword)) {
            displayBookInfo();
        } else {
            System.out.println("Book not found");
        }
    }
}

public class LibrarySystem {
    public static void main(String[] args) {

        Book b = new Book();

        b.bookID = 1;
        b.title = "Java Programming";
        b.author = "James Gosling";

        b.displayBookInfo();

        b.issueBook();
        b.displayBookInfo();

        b.returnBook();
        b.searchBook("Java");
    }
}

```

4. Hotel Room Book System

```

class Room {

    int roomNo;
    String type;
    int rentPerDay;
    boolean isBooked = false;
    String customerName;
    int daysBooked;

    void bookRoom(String name, int days) {

        if (!isBooked) {
            isBooked = true;
            customerName = name;
            daysBooked = days;

            System.out.println("Room Booked Successfully");
        } else {
            System.out.println("Room already booked");
        }
    }

    void vacateRoom() {

```

```

        isBooked = false;
        customerName = "";
        daysBooked = 0;

        System.out.println("Room Vacated");
    }

    int calculateBill() {
        return rentPerDay * daysBooked;
    }

    void displayRoomInfo() {

        System.out.println("Room No: " + roomNo);
        System.out.println("Type: " + type);
        System.out.println("Rent Per Day: " + rentPerDay);

        if (isBooked) {
            System.out.println("Booked By: " + customerName);
            System.out.println("Days: " + daysBooked);
            System.out.println("Total Bill: " + calculateBill());
        } else {
            System.out.println("Room Available");
        }
    }
}

public class HotelBooking {
    public static void main(String[] args) {

        Room r = new Room();

        r.roomNo = 101;
        r.type = "AC";
        r.rentPerDay = 2000;

        r.displayRoomInfo();

        r.bookRoom("Nadeer", 3);

        r.displayRoomInfo();

        r.vacateRoom();
    }
}

```